Note that security was not part of the standard at that time. With increased digitization and adoption of online activities came rampant online fraud and security breaches. Security became critical for web applications, and it was retrofitted into HTTP as a standard (HTTPS) around 2000. This addition enabled secure web applications. Global standards were established with formal Request for Comments (RFC) documents from the Internet Engineering Task Force (IETF)—RFCs 7230, 2818, and so on. Blockchain, introduced in 2009, established a trust layer alongside the security layer of the internet. Trust is currently realized in centralized systems by ad hoc means (such as verifying credentials, recommendation systems, and reviews/ratings) and by human involvement in other situations, such as at airports and grocery-store checkouts. Blockchain enables the trust layer for Dapps through software-based verification, validation, and immutable recording of transactions and facts.

Next, let’s look at some compelling decentralized scenarios that can benefit from the blockchain’s DLT and its trust layer.

1.4 Motivating scenarios

In this section, you explore several issues that are prevalent in the systems you may deal with in everyday activities. Consider the broad area of budgets and expense management in organizations small and large: governmental and nongovernmental agencies (NGOs), charities and disaster-relief agencies, and more. A significant issue is